

QUANTITATIVE RESEARCH — MACRO — CASE STUDY

Forecasting Japanese Inflation to Trade USD/JPY

Constructing an investable currency signal from core-CPI forecasts

Data → Forecast → Strategy → Backtest

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July 2026

Outline

① Introduction

② Methodology

③ Results

④ Conclusion

Japanese macroeconomic view

- ~ 25 years of deflation, near-zero rates
- Jul 2026: weakest yen vs USD since 1986
- Rising government debt
- Services-driven economy
- Import-dependent
- March 2024: policy regime change
 - Exit from NIPR and YCC; first hike in 17 years (Bank of Japan, 2024)
 - Normalization to a 31-year-high rate by 2026; inflation above the 2% target
 - BOJ's gate: wage-sustained inflation — three *Shunto* rounds above 5% (2024–2026)

Trade Idea

- Forecast core-CPI inflation; trade the *surprise* via USD/JPY
- Chain: inflation surprise $\rightarrow$ BOJ repricing $\rightarrow$ JP-US rate differential $\rightarrow$ yen
- Hot surprise $\rightarrow$ short USD/JPY; cool $\rightarrow$ long
- USD/JPY: most liquid, fully reproducible expression
- Regime caveat
 - Sign assumes the post-2024 “BOJ reacts to inflation” regime
 - 2022–2023: inflation up, BOJ dovish $\rightarrow$ yen *weakened* (opposite sign)
 - Bet: reacting regime persists; 2022 is the stress test

Data

- Target: core CPI (“all items, less fresh food”) — BOJ policy measure (Bank of Japan, 2026)
- Sources:
 - National + Tokyo CPI — e-Stat API (Statistics Bureau of Japan, 2026b)
 - USD/JPY, Brent crude — FRED (Federal Reserve Bank of St. Louis, 2026b,a)
- Growth rates, not levels — robust to the 2020→2025 rebase
- Engine: Tokyo leads the nation
 - Tokyo core CPI (month M) prints ~ 3 weeks before the national figure, same month (Statistics Bureau of Japan, 2026a)
- Availability date $\neq$ reference month
- No look-ahead: asserted in code (23-day gap)
- USD/JPY lagged — traded asset, avoids circularity

Forecast model

One-month-ahead, single-equation OLS:

$$\begin{aligned}\text{CPI}_M^{\text{MoM}} \sim & \text{const} + \sum_{k=1}^3 \text{CPI}_{M-k}^{\text{MoM}} \\ & + \text{Tokyo}_M^{\text{MoM}} \\ & + \text{USDJPY}_{M-1}^{\text{ret}} + \text{Oil}_{M-1}^{\text{ret}}\end{aligned}$$

- Not VAR/VARMA: would forecast oil & FX (near random walks)
- Not RNN: too few points, overfits
- OLS: interpretable coefficients

- Honest evaluation
 - Forecast MoM, then reconstruct YoY
 - Avoids YoY persistence flattering a random-walk benchmark (Atkeson; Ohanian, 2001)
 - Expanding-window, out-of-sample
 - Benchmarks: random walk (RW), historical mean (HM)

Strategy

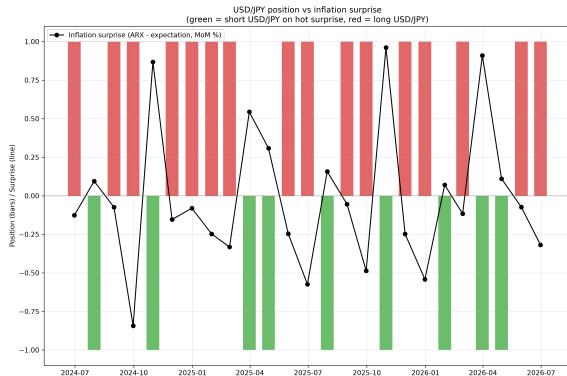


Figure: Monthly inflation surprise and implied USD/JPY position (green: short USD/JPY on a hot surprise; red: long).

- Signal rules

$$\text{surprise}_M = \text{forecast}_M - \text{expectation}_M$$

$$\text{position}_M = -\text{sign}(\text{surprise}_M)$$

- Expectation: naive RW print (last MoM)
- Hot $\rightarrow$ short USD/JPY; cool $\rightarrow$ long; dead-band $\rightarrow$ flat
- Rebalance at forecast date ($\sim$ Tokyo release); replaced next month
- Sizing: unit $\{-1, 0, +1\}$; vol-targeted variant

Result — Forecast against naive benchmarks

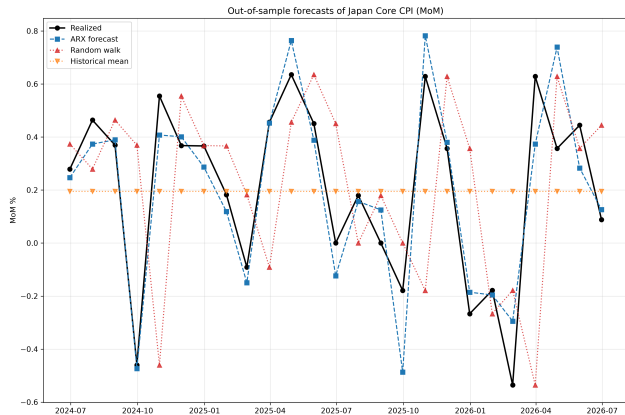


Figure: Out-of-sample 1-month-ahead core-CPI (MoM): ARX vs. realized, random walk and historical mean.

Table: Out-of-sample errors (MoM), test from 2024-06.

	RMSE	MAE
ARX	0.144	0.107
RW	0.464	0.352
HM	0.331	0.281

- ARX RMSE: -69% vs. RW, -56% vs. HM
- YoY (reconstructed): -49% vs. YoY-RW
- Skill from the Tokyo lead — Tokyo prints first

Result — Strategy returns

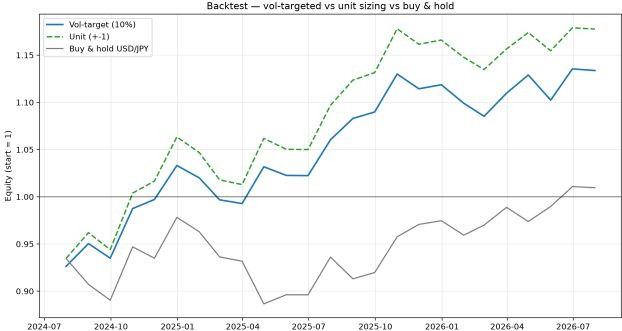


Figure: Net-of-cost equity: unit and vol-targeted strategies vs. buy-and-hold USD/JPY.

Table: Backtest 2024-07 to 2026-07
($n = 25$), net of 1 bp/unit cost, $r_f = 0$.

	Sharpe	Ret.	MaxDD
Unit	0.85	+17.8%	−4.7%
Vol-target	0.69	+13.4%	−4.0%
Buy & hold	0.10	+1.0%	−9.4%

- No look-ahead: position lagged one month; costs each rebalance
- Robustness: drop best month → Sharpe 0.85 → 0.60

Conclusions

- Takeaways
 - Simple, honest, end-to-end pipeline: data $\rightarrow$ forecast $\rightarrow$ signal $\rightarrow$ backtest
 - ARX beats random-walk and mean benchmarks out-of-sample
 - One-instrument USD/JPY signal: Sharpe 0.85 vs. 0.10 buy-and-hold, half the drawdown
- Limitations
 - Regime dependence: sign assumes the post-2024 BOJ (breaks in 2022)
 - Short window (~ 25 months): noisy Sharpe
 - Surprise vs. a naive proxy, not consensus
 - Core CPI still holds energy; carry ignored
- Next steps
 - 2022 regime split
 - Core-core / services signal
 - Consensus surprise
 - Rates instrument

References

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